A UNIT-CUBE BOX-VISIBILITY REPRESENTATION OF K_9

Abstract. We give explicit coordinates for nine pairwise disjoint axis-parallel unit cubes whose box-visibility graph is K_9. This refutes Conjecture 26 of Fekete and Meijer. Together with their theorem that K_{10} has no such representation, this determines 9 as the maximum order of a complete graph representable by unit cubes.

A box-visibility representation assigns pairwise disjoint axis-parallel boxes in $\mathbb{R}^3$ to the vertices of a graph, with two vertices adjacent exactly when their boxes can be joined by an axis-parallel cylindrical channel of positive radius whose ends lie in the box interiors and which meets no other box. Fekete and Meijer constructed such a representation of K_8 by unit cubes, proved that none exists for K_{10}, and conjectured the same for K_9 [1, Fig. 5, Theorem 25, and Conjecture 26].

Theorem 1. *There exist nine pairwise disjoint axis-parallel unit cubes whose box-visibility graph is K_9. In particular, Conjecture 26 of Fekete and Meijer is false.*

Proof. It is convenient to work at scale 16. For $i = 0, \dots, 8$, let

$$C_i = [x_i, x_i + 16] \times [y_i, y_i + 16] \times [z_i, z_i + 16],$$

where

(1)

i	0	1	2	3	4	5	6	7	8
x_i	0	9	-8	10	6	2	7	-5	5
y_i	0	2	3	-1	-7	11	-6	8	1
z_i	0	17	19	34	51	53	68	85	102.

The z-intervals overlap only for the pairs $\{1,2\},\{2,3\},\{4,5\},\{5,6\}$. The first two pairs are strictly separated in the x-direction, and the last two in the y-direction. Thus the cubes are pairwise disjoint. All minima and maxima are pairwise distinct in each coordinate direction.

Write

$$\ell_x(a,b) = \{(t,a,b) : t \in \mathbb{R}\},$$
$$\ell_y(a,b) = \{(a,t,b) : t \in \mathbb{R}\},$$
$$\ell_z(a,b) = \{(a,b,t) : t \in \mathbb{R}\}.$$

The following table gives a visibility axis for every unordered pair; the label $ij : d(a,b)$ means the line $\ell_d(a,b)$ for the pair $\{C_i, C_j\}$.

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$01 : z(19/2, 5/2)$	$02 : z(1/2, 7/2)$	$03 : z(21/2, 1/2)$
$04 : z(13/2, 1/2)$	$05 : z(17/2, 23/2)$	$06 : z(17/2, 19/2)$
$07 : z(17/2, 21/2)$	$08 : z(11/2, 3/2)$	$12 : x(7/2, 39/2)$
$13 : z(21/2, 5/2)$	$14 : z(19/2, 5/2)$	$15 : z(19/2, 23/2)$
$16 : z(19/2, 19/2)$	$17 : z(19/2, 21/2)$	$18 : z(37/2, 31/2)$
$23 : x(7/2, 69/2)$	$24 : z(13/2, 7/2)$	$25 : z(5/2, 23/2)$
$26 : z(15/2, 19/2)$	$27 : z(-9/2, 17/2)$	$28 : z(11/2, 7/2)$
$34 : z(21/2, -1/2)$	$35 : z(21/2, 23/2)$	$36 : z(21/2, 19/2)$
$37 : z(21/2, 21/2)$	$38 : z(23/2, 21/2)$	$45 : y(13/2, 107/2)$
$46 : z(15/2, -11/2)$	$47 : z(13/2, 17/2)$	$48 : z(13/2, 3/2)$
$56 : y(15/2, 137/2)$	$57 : z(5/2, 23/2)$	$58 : z(23/2, 23/2)$
$67 : z(15/2, 17/2)$	$68 : z(15/2, 3/2)$	$78 : z(11/2, 17/2)$

For $d \in \{x, y, z\}$, write p_d for the d-coordinate of $p \in \mathbb{R}^3$, and let π_d denote projection onto the two coordinates transverse to d, ordered as in the definitions of ℓ_d. Fix an entry $ij : d(a,b)$ and put $q = (a,b)$. After exchanging i and j if necessary, let u be the upper d-face of the lower cube and v the lower d-face of the upper cube. Direct substitution in (1) gives, for every table entry,

$$\operatorname{dist}_\infty\Big(q, \mathbb{R}^2 \setminus \operatorname{int}(\pi_d(C_h))\Big) \geq \frac{1}{2} \quad (h \in \{i,j\}),$$
$$\operatorname{dist}_\infty(q, \pi_d(C_k)) \geq \frac{1}{2} \quad \Big(k \notin \{i,j\},\ I_k^d \cap [u - \tfrac{1}{4}, v + \tfrac{1}{4}] \neq \varnothing\Big),$$

where I_k^d is the d-interval of C_k. These are exact finite checks: the face coordinates are integers and the listed transverse coordinates are half-integers. Consequently the cylinder

$$\Gamma_{ij} = \Big\{p \in \mathbb{R}^3 : u - \tfrac{1}{4} \leq p_d \leq v + \tfrac{1}{4},\ \|\pi_d(p) - q\|_2 \leq \tfrac{1}{4}\Big\}$$

has its end disks in the interiors of C_i and C_j and is disjoint from every other cube. Thus every pair of cubes is visible. Dividing all coordinates by 16 gives unit cubes and visibility channels of radius $1/64$. □

The coordinates were found computationally, but the displayed certificate and its verification are exact.

Corollary 2. *The largest n for which K_n has a box-visibility representation by axis-parallel unit cubes is $n = 9$.*

Proof. Theorem 1 gives the lower bound. Fekete and Meijer proved that K_{10} has no such representation [1, Theorem 25]. A representation of any K_n with $n > 10$, restricted to any ten cubes, would give a representation of K_{10}, a contradiction. □

References

[1] S. P. Fekete and H. Meijer, *Rectangle and box visibility graphs in 3D*, Internat. J. Comput. Geom. Appl. **9** (1999), no. 1, 1–27, doi:10.1142/S0218195999000029.